

❖ Repetition Control Instructions (Loop)

- Repetition Control Instructions (Loops) are used to execute a block of code repeatedly until a specified condition becomes false.

➤ 2 Main Types of Loops in Python

Loop	Purpose
while	Executes while a condition is True
for	Iterates over a sequence (list, string, tuple, range, etc.)

1. while Loop

- The while loop repeatedly executes a block of code as long as the specified condition is True.

```
i = 1  
  
while i <= 5:  
    print(i)  
    i += 1
```

2. for Loop

- The for loop is used to iterate over a sequence such as a string, list, tuple, dictionary, set, or range.

```
for i in range(1,6):  
    print(i)
```

```
fruits = ["Apple", "Mango", "Banana"]  
  
for fruit in fruits:  
    print(fruit)
```

Difference Between while and for

while	for
Condition based	Sequence based
Used when number of iterations is unknown	Used when number of iterations is known
Can become infinite if condition is not updated	Less chance of infinite loop

3. range()

- The range() function generates a sequence of numbers. It is commonly used with the for loop.

```
for i in range(5):  
    print(i)
```

4. Nested Loops

- A Nested Loop is a loop inside another loop.

```
for i in range(3):  
    for j in range(2):  
        print(i,j)
```

5. break Statement

- The break statement immediately terminates the loop, regardless of the remaining iterations.

```
for i in range(1,11):  
    if i == 6:  
        break  
    print(i)
```

6. continue Statement

- The continue statement skips the current iteration and moves to the next iteration of the loop.

```
for i in range(1,6):  
  
    if i == 3:  
        continue  
  
    print(i)
```

7. pass Statement

- The pass statement is a placeholder that does nothing. It is used when a statement is syntactically required but no code needs to be executed.

```
for i in range(5):  
  
    if i == 2:  
        pass  
  
    print(i)
```